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Areas of physics by complexity



Newton's  
Mechanics

Electro-  
Magnetism

Special  
Relativity

Quantum Mechanics  
General Relativity

Quantum  
Field Theory

Complexity  
Science

## PoCN Projects

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# 1 | Task #8 - De Domenico-Arenas model

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## 1.1 | Introduction

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The aim is to reproduce the principal findings of De Domenico and Arenas [1] for the Darknet. We will model a synthetic network using their same proposed mechanism and compare it with their empirical January 2015 Tor snapshot, which contains  $N = 5535$  nodes and  $E = 274831$  undirected edges. The main features of the DarkNet that we want to reproduce are:

- **Heavy-tailed degree distribution:**  $P(k)$  heterogeneous and compatible with a truncated power law with scaling exponent equal to  $-1$  and with a cutoff at  $\ln k \approx 6$ .
- **High mean degree:**  $\langle k \rangle = 2E/N \simeq 99.31$ .
- **Clustering and characteristic length:** The local clustering coefficient  $c_i = 2T_i/[k_i(k_i - 1)]$ , where  $T_i$  is the number of closed triangles connected to node  $i$ , is broadly distributed and peaks near 0.3, with an average value of  $\bar{c}_i \approx 0.4$ . The global clustering, calculated as the ratio between the total number of closed triplets and the total number of connected triplets in the network, has a value of  $C \approx 0.2$ . The average path length is about  $l \approx 2.5$  and the diameter is  $D \approx 5$ .
- **Higher-order correlations:**  $k_{nn}(k)$  and  $c(k)$  vary with degree, both have a decreasing behaviour for large  $k$ , but the first is more pronounced. The degree assortativity coefficient  $r$  is around  $-0.25$ . This indicates that hubs tend slightly to connect to lower-degree nodes.
- **No rich-club core.** The normalized rich-club ratio  $\phi(k)/\tilde{\phi}(k)$  stays almost constant near 1, rather than showing a dense core of high-degree nodes.

## 1.2 | Age-bandwidth growth model

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The mechanism used to model the DarkNet is based on age and bandwidth. Each node  $i$  has arrival time  $T_i$ , age  $A_i(\tau) = \tau - T_i$ , and a fixed bandwidth  $B_i$  drawn from a log-normal distribution with parameters  $(\mu, \sigma) = (0, 1)$ . At each step  $\tau$ , a new node  $i$  and  $M$  links enter in the network. The  $2M$  endpoints are chosen randomly with probability

$$p_i(\tau) = \frac{A_i(\tau)^\beta B_i^\gamma}{\sum_j A_j(\tau)^\beta B_j^\gamma}.$$

Here we use the same parameters  $\beta = \gamma = 1$ . This mechanism ensures that new nodes have to increase their age before being linked, in fact in this kind of network we can assume that reputation comes with aging, rather than degree, as classic preferential-attachment networks.

After that, these  $2M$  sampled endpoints are randomly paired, and the last step consists in rewiring nodes randomly while preserving the degree distribution, in order to eliminate possible structural correlations.

The initial network we start with has  $n_0 = 200$  nodes and  $e_0 = 1000$  edges, then we let it grow until it reaches the same size of the 2015 empirical network, that is  $N = 5535$  and  $E = 274831$ .

We generate 50 independent stochastic realizations of the network for calculating the degree distribution and 400 for the structural averages. Moreover, we generated also 1000 random graphs preserving the degree distribution, which served as null models for the normalized rich-club test.

After the network generation, isolated nodes are retained for degree-based metrics, while path-dependent quantities are computed on the largest connected component (LCC). The average path length is estimated in a computationally efficient way by selecting 256 nodes uniformly at random from the LCC. For each sampled node, shortest-path distances to all other LCC nodes are computed and then averaged. The reported diameter is therefore the largest distance observed from those sampled sources (and could be an underestimate of the real diameter). Degree assortativity is computed as the Pearson correlation coefficient between the degrees of the two endpoints of every edge. A negative value indicates that high-degree nodes tend to connect to lower-degree nodes. For the rich-club analysis, nodes with degree greater than a threshold  $k$  are selected and their internal edge density is computed as  $\phi(k) = 2E_{>k} / [N_{>k}(N_{>k} - 1)]$ . The plotted ratio is  $\phi(k)/\tilde{\phi}(k)$ , where  $\tilde{\phi}(k)$  is the expected coefficient in the random network ensemble.

### 1.3 | Results

Table 1.1 (appendix) shows a good agreement between the synthetic network and the empirical 2015 network. The mean degree is the same by construction (it depends just on  $N$  and  $E$ ), while global clustering, mean path length, and diameter are close to the reference values. The synthetic graph has a lower average local clustering and a less negative assortativity coefficient, but it reproduces the same qualitative signatures: substantial clustering and disassortative mixing.

Figure 1.2 (appendix) shows the main characteristics of our synthetic network. Panel (a) presents the degree probability distribution, which is compatible with a power law with scaling exponent  $-1$  and a cutoff around  $\ln k = 6$ , as in the reference paper. Panel (b) shows that the normalized rich-club ratio is close to one over almost all degree thresholds. Hubs are therefore not connected to each other more often than expected from random fluctuations. Panel (c) shows that  $k_{nn}(k)$  decreases markedly in the high-degree tail: the largest hubs are, on average, attached to nodes of lower degree than the intermediate-degree nodes, which is consistent with the negative assortativity reported above. Finally, panel (d) shows that  $c(k)$  remains near 0.3 at moderate degree but decreases slightly for large degrees  $k$ .

### 1.4 | Resilience test

The resilience analysis follows two distinct methodologies, where we test first static failures and then dynamical overload cascades. In both cases the observable is the relative size of the largest connected component (LCC): a small LCC means that the surviving nodes are fragmented and can no longer provide a connected network.

Four different strategies for the static failure test are implemented: random failure and three types of targeted attacks. In random failure, a fraction  $p_{\text{fail}}$  of nodes is selected randomly and removed. For targeted attacks, nodes are instead ranked and removed once by degree, then by betweenness centrality, and finally by  $k$ -coreness. The first targets well-connected nodes, the second targets nodes important for shortest-path traffic, and the third looks at nodes in the innermost shells of the core decomposition, testing how dependent is the network on its structurally deepest core. We record  $S(p_{\text{fail}})/N$ , the relative LCC size, as a function of the surviving-node fraction  $1 - p_{\text{fail}}$ .

The cascade experiment uses the Motter–Lai model. Each node  $i$  has a fixed capacity  $C_i$  and a dynamic load  $L_i(\tau)$ . The load is estimated as the betweenness centrality of node  $i$ : it measures how often shortest paths traverse that node, and it is recomputed on the surviving network after each generation. If the new load exceeds a node’s capacity, that node fails and may trigger a further redistribution in the network. The initial attack removes the highest-degree node of the LCC. We set the baseline capacity to the initial load,  $C_i = L_i(0)$ , and scale it to  $(1 + \alpha)C_i$ , where  $\alpha$  is the capacity-tolerance parameter. At every generation all nodes satisfying  $L_i > (1 + \alpha)C_i$  are removed. The process ends when no overload remains. Each cascade uses a fixed cohort of 128 source nodes to estimate node betweenness centralities. In practice, for each source, shortest paths to all surviving nodes are considered, and the load of a node  $i$  is related to how often it lies on these paths. The correct procedure would be to calculate the betweenness by looping over all the possible pairs of nodes in the network, but we avoided this procedure to gain more computational efficiency.

Therefore, we repeat the experiment for three synthetic-network realizations and three independent cohorts per realization. The final result reports only the mean final LCC over these nine runs.

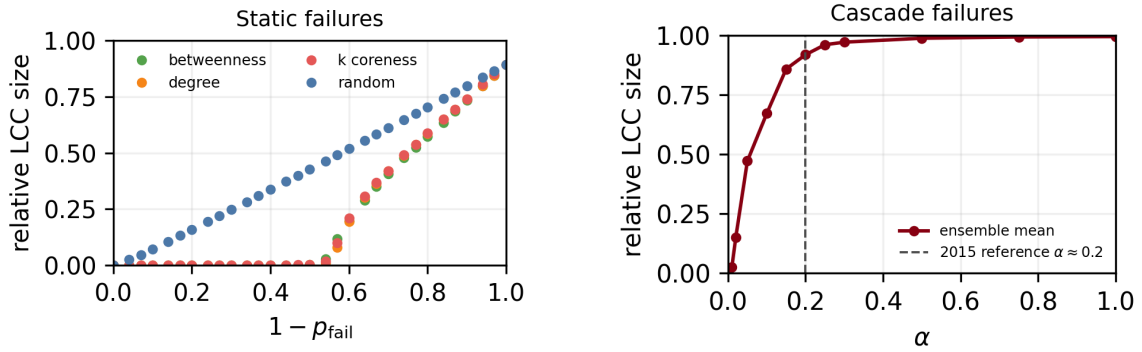


Figure 1.1: Synthetic-network resilience results.

As regards static failures, fig. 1.1 shows that the synthetic graph is more vulnerable to targeted than to random removal, but its targeted fragmentation occurs only near  $p_{\text{fail}} \simeq 0.45$ , equivalently  $1 - p_{\text{fail}} \simeq 0.55$ . This is qualitatively consistent with the empirical reference result  $1 - p_{\text{fail}} \simeq 0.6$ . Instead, for cascade failures, fig. 1.1 reports the final LCC fraction over a sweep of 12 values of  $\alpha$ . Our synthetic network remains fully operative after  $\alpha \simeq 0.3$ , which is in good agreement with the empirical reference  $\alpha \simeq 0.2$ . Yet, even for lower values of  $\alpha$  the network is still not too much fragmented, for example the LCC relative size is 0.5 for  $\alpha = 0.05$ .

## Appendix: Supplementary Darknet results

| Observable               | Synthetic model | Empirical 2015 reference |
|--------------------------|-----------------|--------------------------|
| Mean degree              | 99.31           | 99.31                    |
| Average local clustering | 0.248           | 0.4                      |
| Global clustering        | 0.212           | 0.2                      |
| Mean path length (LCC)   | 2.4             | 2.5                      |
| Diameter                 | 5               | 5                        |
| Degree assortativity     | -0.154          | -0.25                    |

Table 1.1: First synthetic realization versus the 2015 empirical reference.

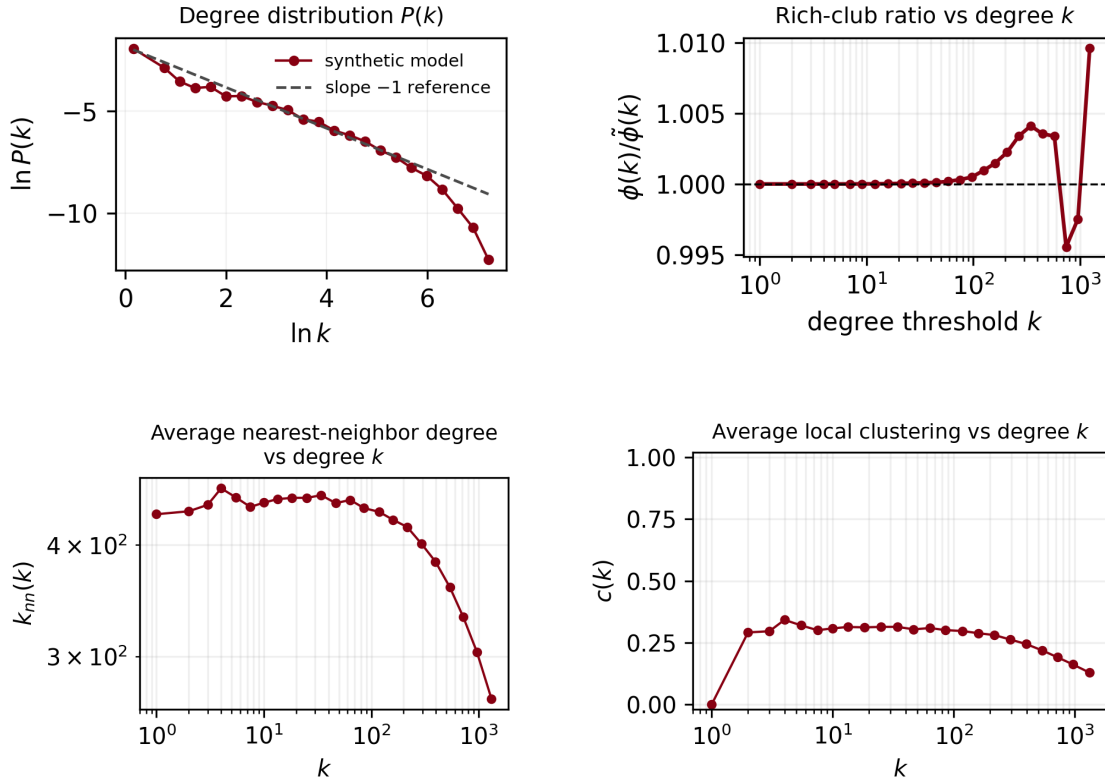


Figure 1.2: Structural and higher-order signatures of the synthetic model.

## 2 | Task #39 – Epidemic spreading on cruise-ship temporal contact networks

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### 2.1 | Introduction

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This project uses cruise ship contact data to investigate how the timing and structure of human interactions affect disease spreading. It compares the empirical time-varying network with its static aggregation and with two randomized surrogate networks, to determine whether neglecting chronological order or empirical contact structure inflates or reduces the predicted size of an outbreak.

Data come from the public repository associated with Pung *et al.* [2]. The two primary files are `dataNodes.RData`, which contains individual-level node metadata, and `dataContact.RData`, which contains contact episodes between pairs of individuals. This analysis uses sail 1 only, for which contacts are observed over three days. We export those data to CSV and construct three processed files: `nodes_sail_1.csv`, containing nodes IDs and additional metadata; `temporal_edges_sail_1.csv`, containing contacts happening all over the three observed ddays with cumulative contact duration; and `aggregated_edges_sail_1.csv`, a static edge list whose weights are total contact durations.

Four network representations are used for the analysis. The *empirical temporal* network preserves the observed interaction day of each pair and uses cumulative pair duration within that day as edge weight. The *aggregated static* network is constructed by aggregating all time-ordered interactions into permanent and simultaneous connections, where the weight of each edge is the total contact duration between the two individuals, divided by the total number of observed days. Two surrogate networks are also used as null models: the *temporal Erdős–Rényi (ER)* network considers the daily empirical edges and preserves the exact number of connections and their weights at each temporal step, but randomly shuffles the interacting nodes; the *static ER* network preserves the number of nodes and edges in the aggregated network, but completely randomizes the connections between nodes and randomly shuffles the existing contact weights among the newly created edges. The reconstructed population (see appendix table 2.1 for the details) contains therefore  $N = 2240$  individuals,  $E = 357\,656$  total temporal contacts among the three days and  $E = 309\,139$  edges in the aggregated static graph. The aggregated network merges repeated contacts between any pair of individual among the three days into a single permanent edge, resulting in a lower count of total edges.

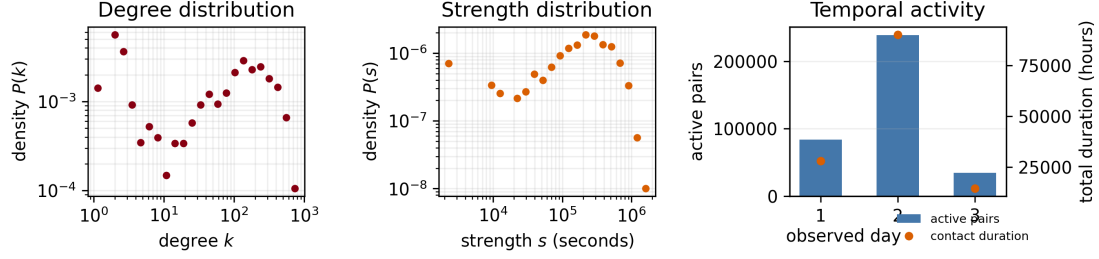


Figure 2.1: Descriptive properties of the empirical sail-1 data. Degree and strength are computed on the aggregated empirical graph; strength is the total contact duration accumulated over the three observed days. The temporal-activity panel shows the number of active pairs and the total contact duration in each daily snapshot.

## 2.2 Epidemic model

The simulation uses a stochastic discrete-time SEIR process. Individuals are in four possible states: susceptible ( $S$ ), exposed ( $E$ ), infectious ( $I$ ), or recovered ( $R$ ). At each time step, infectious individuals can infect susceptible neighbours along the currently active edges. For an edge with duration or weight  $w_{uv}$ , the infection probability is

$$p_{uv} = 1 - \exp\left(-\beta \frac{w_{uv}}{1800}\right)$$

where  $w_{uv}$  is contact duration in seconds and 1800 is the default half-hour scale. Exposed individuals become infectious with probability  $\sigma$  per day, and infectious individuals recover with probability  $\gamma$  per day. The default values are  $\sigma = 0.5$  and  $\gamma = 1/3$ . Since empirical data cover three observed days, the simulation cycles the temporal pattern over a total of 21 days.

The order parameter is the mean attack rate, which is the final fraction of individuals that have left the susceptible state, as function of the transmission probability  $\beta$ . In a single stochastic realization  $r$  it is given by

$$A_r(\beta) = \frac{N_E(T) + N_I(T) + N_R(T)}{N} = 1 - \frac{N_S(T)}{N},$$

where  $T$  is the final simulated day. For  $R$  realizations, the reported order parameter is  $\bar{A}(\beta) = R^{-1} \sum_{r=1}^R A_r(\beta)$ .

## 2.3 Simulation results

For each  $\beta \in \{0.02, 0.05, 0.08, 0.12, 0.16, 0.22, 0.30, 0.40\}$ , the reported mean attack rate is estimated from 1000 independent stochastic realizations. Fig. 2.2 shows that the aggregated static representation systematically gives larger mean attack rates than the empirical temporal network, especially for low  $\beta$  values. This overestimation decreases for large  $\beta$  and the reason is that, when  $\beta \rightarrow 1$ , almost every contact leads to transmission, therefore after 21 days of simulation all the networks should exhibit

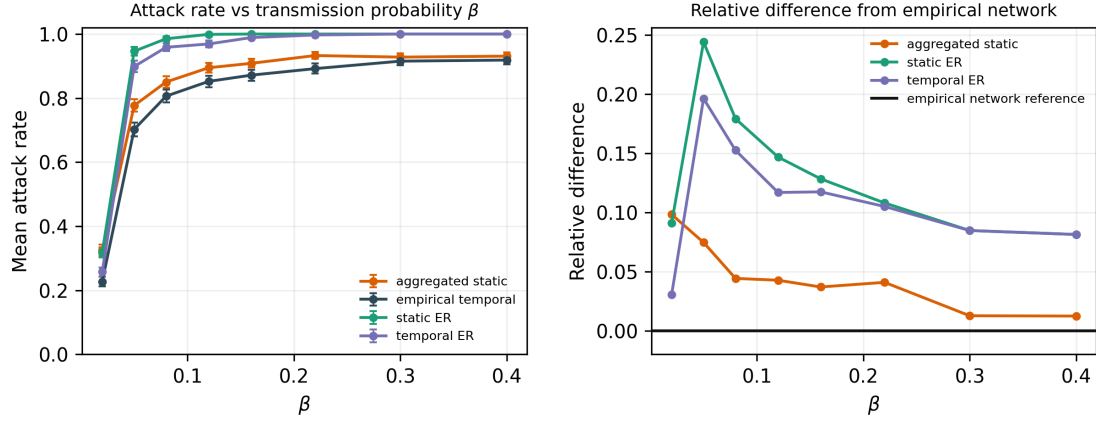


Figure 2.2: SEIR results for the four network representations. Left panel: mean attack rate as function of the transmission probability  $\beta$  across the four different network representations. Right panel: relative difference between each network attack rate and the empirical temporal network attack rate. The horizontal black thick line refers to the temporal network reference.

an attack rate close to 1. Instead, at intermediate  $\beta$  values, chronology of contacts matters much more and leads to differences in the results. The ER surrogate network show even stronger overestimations for all  $\beta$  values. In fact, temporal ER shuffles the contact pairs and thus removes correlations between who meets whom, resulting in an epidemic that reaches different groups more efficiently. On the other hand, static ER randomizes the topology as well, but it also makes contacts permanently available, producing the largest outbreaks of the epidemic.

The conclusion is that, for this cruise population, replacing the specific temporal sequence of contacts with links that are permanently available inflates final epidemic size, particularly at moderate transmission. The comparison of the surrogate ER networks with the empirical representations shows that the larger outbreaks on both ER networks indicate that the observed organization of contacts, and not only their number and duration, limits epidemic spreading.

## Appendix: Network characteristics and seed-node dependence optional extension

| Representation     | Nodes | Edges                     | Mean degree         | Average local clustering $\bar{C}$ | Degree assortativity $r$ |
|--------------------|-------|---------------------------|---------------------|------------------------------------|--------------------------|
| Empirical temporal | 2240  | (83 759, 239 279, 34 618) | (74.8, 213.6, 30.9) | 0.274                              | 0.154                    |
| Aggregated static  | 2240  | 309 139                   | 276.02              | 0.321                              | 0.191                    |
| Temporal ER        | 2240  | (83 759, 239 279, 34 618) | (74.8, 213.6, 30.9) | 0.048                              | -0.003                   |
| Static ER          | 2240  | 309 139                   | 276.02              | 0.123                              | -0.001                   |

Table 2.1: Network characteristics used in the simulations. For temporal networks, clustering and assortativity are averaged across the three daily snapshots; for static networks, their values are computed on the single static graph. Both metrics use the unweighted topology. This means that, for example, the clustering just counts the number of triangles without considering the duration of the contacts.

### Dependence on the initial seed node

The seed-node experiment samples the initial infectious node uniformly from one of five pools: all nodes (random), passengers, crew members, the top 10% of nodes by aggregated degree, or the bottom 10% of degree nodes. The same eight transmission values  $\beta$ , 21-day horizon, and SEIR parameters are retained; each point is the mean over 300 independent realizations. Only the empirical temporal and aggregated static representations are considered here.

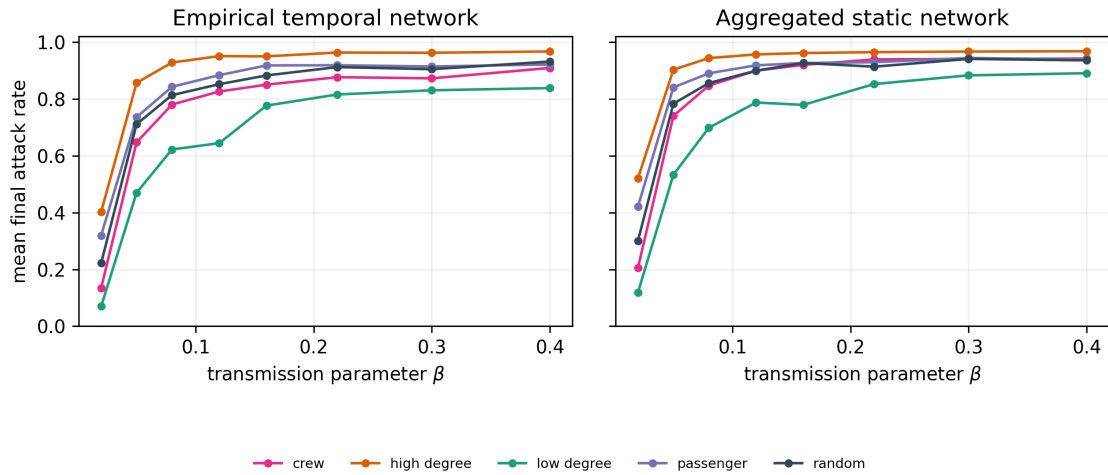


Figure 2.3: Dependence of mean final attack rate on the initial seed-node pool.

The result is that, if the epidemic starts on high-degree nodes, it produces large outbreaks in both representations even for low values of  $\beta$ . On the other hand, initial infected low-degree nodes produce lower final attack rate. Finally, random seed nodes final attack rates are placed between crew and passenger respective attack rates. This does not surprise because there are far more passenger than crew members in the ship,

and a random sample is likely to fall more often in the passenger class rather than the crew one.

The static representation again gives larger outbreaks for the same seed-selection strategy, but the differences are not so high when it comes to high-degree nodes selection strategy.

## 3 | Bibliography

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- [2] Rachael Pung et al. Cruise networks. Zenodo data archive and source repository. URL [https://github.com/rachaelpung/cruise\\_networks](https://github.com/rachaelpung/cruise_networks).